Assignment 2: Alignment and Reasoning RL

1 Assignment Overview

In this assignment, you will gain some hands-on experience with training language models to reason when solving math problems.

What you will implement.

1. Zero-shot prompting baseline for the MATH dataset of competition math problems Hendrycks et al. [2021].
2. Supervised finetuning, given reasoning traces from a stronger reasoning model (DeepSeek R1, DeepSeek-AI et al. 2025).
3. Expert Iteration for improving reasoning performance with verified rewards.
4. Group-Relative Policy Optimization (GRPO) for improving reasoning performance with verified rewards.

What you will run.

1. Measure Qwen 2.5 Math 1.5B zero-shot prompting performance (our baseline).
2. Run SFT on Qwen 2.5 Math 1.5B with reasoning traces from R1.
3. Run Expert Iteration on Qwen 2.5 Math 1.5B with verified rewards.
4. Run GRPO on Qwen 2.5 Math 1.5B with verified rewards.

What the code looks like. As in previous assignments, we use `uv` to manage dependencies. Install all packages except `flash-attn`, then all packages (`flash-attn` is weird):

```
uv sync --no-install-package flash-attn
uv sync
```

1. `alignment/*`: This is where you'll write your code. Note that there's no code in here (aside from a little starter code), so you should be able to do whatever you want from scratch.
2. `alignment/prompts/*`: For your convenience, we've provided text files with prompts to minimize possible errors caused by copying-and-pasting prompts from the PDF to your code.
3. `tests/*.py`: This contains all the tests that you must pass. **You are only expected to pass the tests in `tests/test_sft.py` and `tests/test_grpo.py`—the rest of the tests are for the non-mandatory parts of the assignment.** These tests invoke the hooks defined in `tests/adapters.py`. You'll implement the adapters to connect your code to the tests. Writing more tests and/or modifying the test code can be helpful for debugging your code, but your implementation is expected to pass the original provided test suite.

To run unit tests: `uv run pytest`. Initially, all tests should fail with `NotImplementedError`s. To connect your implementation to the tests, complete the functions in `./tests/adapters.py`.

What you can use. We expect you to build most of the RL related components from scratch. You may use tools like vLLM to generate text from language models (§3.1). In addition, you may use HuggingFace Transformers to load the Qwen 2.5 Math 1.5B model and tokenizer and run forward passes (§4.1), but you may not use any of the training utilities (e.g., the `Trainer` class).

How to submit. You will submit the following files:

- `writeup.pdf`: Answer all the written questions. Please typeset your responses. Submit to Canvas.
- `code.zip`: Contains all the code you've written. Submit to Gradescope.

2 Reasoning with Language Models

2.1 Motivation

One of the remarkable use cases of language models is in building generalist systems that can handle a wide range of natural language processing tasks. In this assignment, we will focus on a developing use case for language models: mathematical reasoning. It will serve as a testbed for us to set up evaluations, perform supervised finetuning, and experiment with teaching LMs to reason using reinforcement learning (RL).

There are going to be two differences from the way we've done our past assignments.

- First, we are not going to be using our language model codebase and models from earlier. We would ideally like to use base language models trained from previous assignments, but finetuning those models will not give us a satisfying result—these models are far too weak to display non-trivial mathematical reasoning capabilities. Because of this, we are going to switch to a modern, high-performance language model that we can access (Qwen 2.5 Math 1.5B Base) and do most of our work on top of that model.
- Second, we are going to introduce a new benchmark with which to evaluate our language models. Up until this point, we have embraced the view that cross-entropy is a good surrogate for many downstream tasks. However, the point of this assignment will be to bridge the gap between base models and downstream tasks and so we will have to use evaluations that are separate from cross-entropy. We will use the MATH 12K dataset from Hendrycks et al. [2021], which consists of challenging high-school competition mathematics problems. We will evaluate language model outputs by comparing them against a reference answer.

2.2 Chain-of-Thought Reasoning and Reasoning RL

An exciting recent trend in language models is the use of *chain-of-thought* reasoning to improve performance across a variety of tasks. Chain-of-thought refers to the process of reasoning through a problem step-by-step, generating intermediate reasoning steps before arriving at a final answer.

Chain-of-thought reasoning with LLMs. Early chain-of-thought approaches finetuned language models to solve simple mathematical tasks like arithmetic by using a "scratchpad" to break the problem into intermediate steps [Nye et al., 2021]. Other work prompts a strong model to "think step by step" before answering, finding that this significantly improves performance on mathematical reasoning tasks like grade-school math questions [Wei et al., 2023].

Learning to reason with expert iteration. The Self-Taught Reasoner (STaR) [Zelikman et al., 2022] frames reasoning as a bootstrapping loop: a pretrained model first samples diverse chains-of-thought (CoTs), keeps only those that lead to correct answers, and then finetunes on these "expert" traces. Iterating this cycle can improve the LM's reasoning capabilities and solve rate. STaR demonstrated that this version of expert iteration [Anthony et al., 2017] using automatic, string match–based verification of generated answers can bootstrap reasoning skills without human-written reasoning traces.

Reasoning RL with verified rewards, o1, and R1. Recent work has explored using more powerful reinforcement learning algorithms with verified rewards to improve reasoning performance. OpenAI's o1 (and subsequent o3/o4) [OpenAI et al., 2024], DeepSeek's R1 [DeepSeek-AI et al., 2025], and Moonshot's kimi k1.5 [Team et al., 2025] use policy gradient methods [Sutton et al., 1999] to train on math and code tasks where string matching or unit tests verify correctness, demonstrating remarkable improvements in competition math and coding performance. Later works such as Open-R1 [Face, 2025], SimpleRL-Zoo [Zeng et al., 2025], and TinyZero [Pan et al., 2025] confirm that pure reinforcement learning with verified rewards—even on models as small as 1.5B parameters—can improve reasoning performance.

Our setup: model and dataset. In the following sections, we will consider progressively more complex approaches to train a base language model to reason step-by-step in order to solve math problems. For this assignment, we will be using the Qwen 2.5 Math 1.5B Base model, which was continually pretrained from the Qwen 2.5 1.5B model on high-quality synthetic math pretraining data [Yang et al., 2024]. The MATH dataset is available at `data/MATH`.

Tip for Open-Source Auditors: Alternative Datasets

Unfortunately, the MATH dataset is not publicly available due to a copyright claim. If you are following along at home, you can use one of the following open-source mathematical reasoning datasets:

- Countdown [Pan et al., 2025], available here: a simple synthetic task based on the British TV show Countdown that has served as a popular testbed for small-scale reasoning RL.
- GSM8K [Cobbe et al., 2021a], available here: grade-school math problems, which are easier than MATH but should allow you to debug correctness and get familiar with the reasoning RL pipeline.
- Tulu 3 SFT Math [Lambert et al., 2025], available here: synthetic math problems generated using GPT-4o and Claude 3.5 Sonnet. Because these are synthetic, some answers (or even the questions) may not be entirely correct.
- Some other math SFT dataset linked here.

To obtain short ground-truth labels (e.g., `1/2`) if they are not provided directly, you can process the ground-truth column with a math answer parser such as Math-Verify.

3 Measuring Zero-Shot MATH Performance

We'll start by measuring the performance of our base language model on the 5K example test set of MATH. Establishing this baseline is useful for understanding how each of the later approaches affects model behavior.

Unless otherwise specified, for experiments on MATH we will use the following prompt from the DeepSeek R1-Zero model [DeepSeek-AI et al., 2025]. We will refer to this as the `r1_zero` prompt:

```
A conversation between User and Assistant. The User asks a question, and the Assistant
↪  solves it. The Assistant first thinks about the reasoning process in the mind and
↪  then provides the User with the answer. The reasoning process is enclosed within
↪  <think> </think> and answer is enclosed within <answer> </answer> tags, respectively,
↪  i.e., <think> reasoning process here </think> <answer> answer here </answer>.
User: {question}
Assistant: <think>
```

The `r1_zero` prompt is located in the text file `alignment/prompts/r1_zero.prompt`.

In the prompt, `question` refers to some question that we insert (e.g., `Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia`

sell altogether in April and May?). The expectation is that the model plays the role of the assistant, and starts generating the thinking process (since we have already included a left think tag <think>), closes the thinking process with </think> and then generates a final symbolic answer within the answer tags, like <answer> 4x + 10 </answer>. The purpose of having the model generate tags like <answer> </answer> is so that we can easily parse the model's output and compare it against a ground truth answer, and so that we can stop response generation when we see the right answer tag </answer>.

Note on prompt choice. It turns out that the r1_zero prompt is not the best choice for maximizing downstream performance after RL, because of a mismatch between the prompt and how the Qwen 2.5 Math 1.5B model was pretrained. Liu et al. [2025] finds that simply prompting the model with the question (and nothing else) starts with a very high accuracy, e.g., matching the r1_zero prompt after 100+ steps of RL. Their findings suggest that Qwen 2.5 Math 1.5B was already pretrained on such question-answer pairs.

Nonetheless, we choose the r1_zero prompt for this assignment because RL with it shows clear accuracy improvements in a short number of steps, allowing us to walk through the mechanics of RL and sanity check correctness quickly, even if we don't manage the best final performance. As a reality check, you will compare directly to the question_only prompt later in the assignment.

3.1 Using vLLM for offline language model inference

To evaluate our language models, we're going to have to generate continuations (responses) for a variety of prompts. While one could certainly implement their own functions for generation (e.g., as you did in assignment 1), efficient implementation of RL requires high-performance inference techniques, and implementing these inference techniques are beyond the scope of this assignment. Therefore, in this assignment we will recommend using vLLM for offline batched inference. vLLM is a high-throughput and memory-efficient inference engine for language models that incorporates a variety of useful efficiency techniques (e.g., optimized CUDA kernels, PagedAttention for efficient attention KV caching [Kwon et al., 2023], etc.). To use vLLM to generate continuations for a list of prompts:[1]

```
from vllm import LLM, SamplingParams

# Sample prompts.
prompts = [
    "Hello, my name is",
    "The president of the United States is",
    "The capital of France is",
    "The future of AI is",
]

# Create a sampling params object, stopping generation on newline.
sampling_params = SamplingParams(
    temperature=1.0, top_p=1.0, max_tokens=1024, stop=["\n"]
)

# Create an LLM.
llm = LLM(model=<path to model>)

# Generate texts from the prompts. The output is a list of RequestOutput objects
# that contain the prompt, generated text, and other information.
outputs = llm.generate(prompts, sampling_params)

# Print the outputs.
```

[1] Example taken from https://github.com/vllm-project/vllm/blob/main/examples/offline_inference.py.

```
for output in outputs:
    prompt = output.prompt
    generated_text = output.outputs[0].text
    print(f"Prompt: {prompt!r}, Generated text: {generated_text!r}")
```

In the example above, the `LLM` can be initialized with the name of a HuggingFace model (which will be automatically downloaded and cached if it isn't found locally), or a path to a HuggingFace model.

- Qwen 2.5 Math 1.5B Base (for reasoning experiments):
  `https://huggingface.co/Qwen/Qwen2.5-Math-1.5B`

3.2 Zero-shot MATH Baseline

Prompting setup. To evaluate zero-shot performance on the MATH test set, we'll simply load the examples and prompt the language model to answer the question using the `r1_zero` prompt from above.

Evaluation metric. When we evaluate a multiple-choice or binary response task, the evaluation metric is clear—we test whether the model outputs exactly the correct answer.

In math problems we assume that there is a known ground truth (e.g. `0.5`) but we cannot simply test whether the model outputs exactly 0.5—it can also answer `<answer> 1/2 </answer>`. Because of this, we must address the tricky problem of matching for semantically equivalent responses from the LM when we evaluate MATH.

To this end, we want to come up with some answer parsing function that takes as input the model's output and a known ground-truth, and returns a boolean indicating whether the model's output is correct. For example, a reward function could receive the model's string output ending in `<answer> She sold 15 clips. </answer>` and the gold answer `72`, and return `True` if the model's output is correct and `False` otherwise (in this case, it should return `False`).

For our MATH experiments, we will use a fast and fairly accurate answer parser used in recent work on reasoning RL [Liu et al., 2025]. This reward function is implemented at `alignment.drgrpo_grader.r1_zero_reward_fn`, and you should use it to evaluate performance on MATH unless otherwise specified.

Generation hyperparameters. When generating responses, we'll sample with temperature 1.0, top-p 1.0, max generation length 1024. The prompt asks the model to end its answer with the string `</answer>`, and therefore we can direct vLLM to stop when the model outputs this string:

```
# Based on Dr. GRPO: stop when the model completes its answer
# https://github.com/sail-sg/understand-r1-zero/blob/
#   c18804602b85da9e88b4aeeb6c43e2f08c594fbc/train_zero_math.py#L167
sampling_params.stop = ["</answer>"]
sampling_params.include_stop_str_in_output = True
```

Problem (`math_baseline`):

(a) Write a script to evaluate Qwen 2.5 Math 1.5B zero-shot performance on MATH. This script should (1) load the MATH validation examples from `data/MATH/validation.jsonl`, (2) format them as string prompts to the language model using the `r1_zero` prompt, and (3) generate outputs for each example. This script should also (4) calculate evaluation metrics and (5) serialize the examples, model generations, and corresponding evaluation scores to disk for analysis in subsequent problems.

It might be helpful for your implementation to include a method `evaluate_vllm` with arguments

similar to the following, as you will be able to reuse it later:

```
def evaluate_vllm(
    vllm_model: LLM,
    reward_fn: Callable[[str, str], dict[str, float]],
    prompts: List[str],
    eval_sampling_params: SamplingParams
) -> None:
    """
    Evaluate a language model on a list of prompts,
    compute evaluation metrics, and serialize results to disk.
    """
```

Deliverable: A script to evaluate baseline zero-shot MATH performance.

(b) Run your evaluation script on Qwen 2.5 Math 1.5B. How many model generations fall into each of the following categories: (1) correct with both format and answer reward 1, (2) format reward 1 and answer reward 0, (3) format reward 0 and answer reward 0? Observing at least 10 cases where format reward is 0, do you think the issue is with the base model's output, or the parser? Why? What about in (at least 10) cases where format reward is 1 but answer reward is 0?

Deliverable: Commentary on the model and reward function performance, including examples of each category.

(c) How well does the Qwen 2.5 Math 1.5B zero-shot baseline perform on MATH?

Deliverable: 1-2 sentences with evaluation metrics.

4 Supervised Finetuning for MATH

Algorithm 1 Supervised Finetuning (SFT)

Input initial policy model $\pi_{\theta_{\text{init}}}$; SFT dataset $\mathcal{D}$

policy model $\pi_\theta \leftarrow \pi_{\theta_{\text{init}}}$
for step = 1, ..., `n_sft_steps` **do**
    Sample a batch of question-response pairs $\mathcal{D}_b$ from $\mathcal{D}$
    Compute the cross-entropy loss of the responses given the questions using the model π_θ
    Update the model parameters θ by taking a gradient step with respect to the cross-entropy loss
end for

Output π_θ

Supervised finetuning for reasoning. In this section, we will finetune our base model on the MATH dataset (Algorithm 1). As our goal is to improve the model's reasoning ability, rather than finetune it to directly predict correct answers, we will finetune it to first generate a chain-of-thought reasoning trace followed by an answer. To this end, we have made available a dataset of such reasoning traces, obtained from DeepSeek R1 DeepSeek-AI et al. [2025], in

`data/sft.jsonl`

When training a reasoning model in practice, SFT is often used as a warm-start for a second RL finetuning step. There are two main reasons for this. First, SFT requires high-quality annotated data (i.e., with pre-existing reasoning traces), whereas RL requires only the correct answer for feedback. Second, even in settings where annotated data is plentiful, RL can still unlock performance gains by finding better policies than the SFT data. Unfortunately, the models we use are not big enough to show effects when composing SFT and RL, so for this assignment we will treat these two phases separately.

4.1 Using HuggingFace Models

Loading a HuggingFace model and tokenizer. To load a HuggingFace model and tokenizer from a local dir (in `bfloat16` and with FlashAttention-2 to save memory), you can use the following starter code:

```
from transformers import AutoModelForCausalLM, AutoTokenizer

model = AutoModelForCausalLM.from_pretrained(
    "models/Qwen2.5-Math-1.5B",
    torch_dtype=torch.bfloat16,
    attn_implementation="flash_attention_2",
)
tokenizer = AutoTokenizer.from_pretrained("models/Qwen2.5-Math-1.5B")
```

Forward pass. After we've loaded the model, we can run a forward pass on a batch of input IDs and get the logits (with the `.logits`) attribute of the output. Then, we can compute the loss between the model's predicted logits and the actual labels:

```
input_ids = train_batch["input_ids"].to(device)
labels = train_batch["labels"].to(device)

logits = model(input_ids).logits
loss = F.cross_entropy(..., ...)
```

Saving a trained model. To save the model to a directory after training is finished, you can use the `.save_pretrained()` function, passing in the path to the desired output directory. We recommend also saving the tokenizer as well (even if you didn't modify it), just so the model and tokenizer are self-contained and loadable from a single directory.

```
# Save the model weights
model.save_pretrained(save_directory=output_dir)
tokenizer.save_pretrained(save_directory=output_dir)
```

Gradient accumulation. Despite loading the model in `bfloat16` and using FlashAttention-2, even an 80GB GPU does not have enough memory to support reasonable batch sizes. To use larger batch sizes, we can use a technique called *gradient accumulation*. The basic idea behind gradient accumulation is that rather than updating our model weights (i.e., taking an optimizer step) after every batch, we'll *accumulate* the gradients over several batches before taking a gradient step. Intuitively, if we had a larger GPU, we should get the same results from computing the gradient on a batch of 32 examples all at once, vs. splitting them up into 16 batches of 2 examples each and then averaging at the end.

Gradient accumulation is straightforward to implement in PyTorch. Recall that each weight tensor has an attribute `.grad` that stores its gradient. Before we call `loss.backward()`, the `.grad` attribute is None. After we call `loss.backward()`, the `.grad` attribute contains the gradient. Normally, we'd take an optimizer step, and then zero the gradients with `optimizer.zero_grad()`, which resets the `.grad` field of the weight tensors:

```
for inputs, labels in data_loader:
    # Forward pass.
    logits = model(inputs)
    loss = loss_fn(logits, labels)

    # Backward pass.
    loss.backward()
```

```
    # Update weights.
    optimizer.step()
    # Zero gradients in preparation for next iteration.
    optimizer.zero_grad()
```

To implement gradient accumulation, we'll just call the `optimizer.step()` and `optimizer.zero_grad()` every k steps, where k is the number of gradient accumulation steps. We divide the loss by `gradient_accumulation_steps` before calling `loss.backward()` so that the gradients are averaged across the gradient accumulation steps.

```
gradient_accumulation_steps = 4
for idx, (inputs, labels) in enumerate(data_loader):
    # Forward pass.
    logits = model(inputs)
    loss = loss_fn(logits, labels) / gradient_accumulation_steps

    # Backward pass.
    loss.backward()

    if (idx + 1) % gradient_accumulation_steps == 0:
        # Update weights every `gradient_accumulation_steps` batches.
        optimizer.step()
        # Zero gradients every `gradient_accumulation_steps` batches.
        optimizer.zero_grad()
```

As a result, our effective batch size when training is multiplied by k, the number of gradient accumulation steps.

4.2 SFT Helper Methods

Next, we will implement some helper methods that you will use during SFT and in the later RL experiments. As a quick note on nomenclature: in the following sections, we will interchangeably refer to a model's completion given a prompt as an "output", "completion", or "response".

Tokenizing prompts and outputs. For each pair of question and target output (q, o), we will tokenize the question and output separately and concatenate them. Then, we can score the log-probabilities of the output with our SFT model (or in later sections, our RL policy). Moreover, we will need to construct a `response_mask`: a boolean mask that is `True` for all tokens in the response, and `False` for all question and padding tokens. We will use this mask in the training loop to ensure that we only compute the loss on the response tokens.

Problem (`tokenize_prompt_and_output`): Prompt and output tokenization

Deliverable: Implement a method `tokenize_prompt_and_output` that tokenizes the question and output separately, concatenates them together, and constructs a `response_mask`. The following interface is recommended:

`def tokenize_prompt_and_output(prompt_strs, output_strs, tokenizer):` Tokenize the prompt and output strings, and construct a mask that is 1 for the response tokens and 0 for other tokens (prompt or padding).

Args:

`prompt_strs: list[str]` List of prompt strings.

`output_strs: list[str]` List of output strings.

`tokenizer: PreTrainedTokenizer` Tokenizer to use for tokenization.

Returns:

`dict[str, torch.Tensor]`. Let `prompt_and_output_lens` be a list containing the lengths of the tokenized prompt and output strings. Then the returned dictionary should have the following keys:

`input_ids` torch.Tensor of shape `(batch_size, max(prompt_and_output_lens) - 1)`: the tokenized prompt and output strings, with the final token sliced off.

`labels` torch.Tensor of shape `(batch_size, max(prompt_and_output_lens) - 1)`: shifted input ids, i.e., the input ids without the first token.

`response_mask` torch.Tensor of shape `(batch_size, max(prompt_and_output_lens) - 1)`: a mask on the response tokens in the labels.

To test your code, implement `[adapters.run_tokenize_prompt_and_output]`. Then, run the test with `uv run pytest -k test_tokenize_prompt_and_output` and make sure your implementation passes it.

Logging per-token entropies. When doing RL, it is often useful to keep track of per-token entropies to see if the predictive distribution of the model is becoming (over)confident. We will implement this now and compare how each of our finetuning approaches affects the model's predictive entropy.

The entropy of a discrete distribution $p(x)$ with support $\mathcal{X}$ is defined as

$$H(p) = -\sum_{x \in \mathcal{X}} p(x) \log p(x). \tag{1}$$

Given our SFT or RL model's logits, we will compute the per-token entropy, i.e., the entropy of each next-token prediction.

Problem (`compute_entropy`): Per-token entropy

Deliverable: Implement a method `compute_entropy` that computes the per-token entropy of next-token predictions.
The following interface is recommended:

```
def compute_entropy(logits: torch.Tensor) -> torch.Tensor:
```

Get the entropy of the next-token predictions (i.e., entropy over the vocabulary dimension).

Args:

`logits: torch.Tensor` Tensor of shape `(batch_size, sequence_length, vocab_size)` containing unnormalized logits.

Returns:

`torch.Tensor` Shape `(batch_size, sequence_length)`. The entropy for each next-token prediction.

Note: you should use a numerically stable method (e.g., using `logsumexp`) to avoid overflow.

To test your code, implement `[adapters.run_compute_entropy]`. Then run `uv run pytest -k test_compute_entropy` and ensure your implementation passes.

Getting log-probabilities from a model. Obtaining log-probabilities from a model is a primitive that we will need in both SFT and RL.

For a prefix x, an LM producing next-token logits $f_\theta(x) \in \mathbb{R}^{|\mathcal{V}|}$, and a label $y \in \mathcal{V}$, the log-probability of y is

$$\log p_\theta(y \mid x) = \log \left[\text{softmax}(f_\theta(x))\right]_y , \tag{2}$$

where the notation $[x]_y$ denotes the y-th element of the vector x.

You will want to use a numerically stable method to compute this, and are free to use methods from `torch.nn.functional`. We also suggest including an argument to optionally compute and return token entropies.

Problem (`get_response_log_probs`): Response log-probs (and entropy)

Deliverable: Implement a method `get_response_log_probs` that gets per-token conditional log-probabilities (given the previous tokens) from a causal language model, and optionally the entropy of the model's next-token distribution.
The following interface is recommended:

```
def get_response_log_probs(
    model: PreTrainedModel,
    input_ids: torch.Tensor,
    labels: torch.Tensor,
    return_token_entropy: bool = False,
) -> dict[str, torch.Tensor]:
```

Args:

`model: PreTrainedModel` HuggingFace model used for scoring (placed on the correct device and in inference mode if gradients should not be computed).

`input_ids: torch.Tensor` shape `(batch_size, sequence_length)`, concatenated prompt + response tokens as produced by your tokenization method.

`labels: torch.Tensor` shape `(batch_size, sequence_length)`, labels as produced by your tokenization method.

`return_token_entropy: bool` If `True`, also return per-token entropy by calling `compute_entropy`.

Returns:

`dict[str, torch.Tensor]`.

`"log_probs"` shape `(batch_size, sequence_length)`, conditional log-probabilities $\log p_\theta(x_t \mid x_{<t})$.

`"token_entropy"` optional, shape `(batch_size, sequence_length)`, per-token entropy for each position (present only if `return_token_entropy=True`).

Implementation tips:

- Obtain logits with `model(input_ids).logits`.

To test your code, implement [adapters.run_get_response_log_probs]. Then run `uv run pytest -k test_get_response_log_probs` and ensure the test passes.

SFT microbatch train step. The loss we minimize in SFT is the negative log-likelihood of the target output given the prompt. To compute this loss, we need to compute the log-probabilities of the target output given the prompt and sum over all tokens in the output, masking the tokens in the prompt and padding tokens.

We will implement a helper function for this, that we will also make use of later during RL.

Problem (`masked_normalize`): Masked normalize

Deliverable: Implement a method `masked_normalize` that sums over tensor elements and normalizes by a constant while respecting a boolean mask.
The following interface is recommended:

```
def masked_normalize(
    tensor: torch.Tensor,
    mask: torch.Tensor,
    normalize_constant: float,
    dim: int | None = None,
) -> torch.Tensor:
```

Sum over a dimension and normalize by a constant, considering only those elements where `mask == 1`.

Args:

`tensor: torch.Tensor` The tensor to sum and normalize.

`mask: torch.Tensor` Same shape as `tensor`; positions with `1` are included in the sum.

`normalize_constant: float` the constant to divide by for normalization.

`dim: int | None` the dimension to sum along before normalization. If None, sum over all dimensions.

Returns:

`torch.Tensor` the normalized sum, where masked elements (`mask == 0`) don't contribute to the sum.

To test your code, implement [adapters.run_masked_normalize]. Then run `uv run pytest -k test_masked_normalize` and ensure it passes.

SFT microbatch train step. We are now ready to implement a single microbatch train step for SFT (recall that for a train minibatch, we iterate over many microbatches if `gradient_accumulation_steps > 1`).

Problem (`sft_microbatch_train_step`): Microbatch train step

Deliverable: Implement a single micro-batch update for SFT, including cross-entropy loss, summing with a mask, and gradient scaling.
The following interface is recommended:

```python
def sft_microbatch_train_step(
    policy_log_probs: torch.Tensor,
    response_mask: torch.Tensor,
    gradient_accumulation_steps: int,
    normalize_constant: float = 1.0,
) -> tuple[torch.Tensor, dict[str, torch.Tensor]]:
```

Execute a forward-and-backward pass on a microbatch.

Args:

`policy_log_probs` `(batch_size, sequence_length)`, per-token log-probabilities from the SFT policy being trained.

`response_mask` `(batch_size, sequence_length)`, `1` for response tokens, `0` for prompt/padding.

`gradient_accumulation_steps` Number of microbatches per optimizer step.

`normalize_constant` The constant by which to divide the sum. It is fine to leave this as 1.0.

Returns:

`tuple[torch.Tensor, dict[str, torch.Tensor]]`.

loss scalar tensor. The microbatch loss, adjusted for gradient accumulation. We return this so we can log it.

metadata Dict with metadata from the underlying loss call, and any other statistics you might want to log.

Implementation tips:

- You should call `loss.backward()` in this function. Make sure to adjust for gradient accumulation.

To test your code, implement `[adapters.run_sft_microbatch_train_step]`. Then run `uv run pytest -k test_sft_microbatch_train_step` and confirm it passes.

Logging generations in-the-loop. It's always good practice to do some in-the-loop logging that involves generation from your model, and reasoning SFT/RL is no exception. Write a function `log_generations` that will prompt your model to generate responses for some given prompts (e.g., sampled from the validation set). It's a good idea to log at least the following for each example:

1. The input prompt.
2. The response generated by the SFT/RL model.
3. The ground-truth answer.
4. The reward information, including format, answer, and total reward.
5. The average token entropy of the response.
6. The average response length, average response length for correct responses, and average response length for incorrect responses.

Problem (`log_generations`): Logging generations

Deliverable: Implement a function `log_generations` that can be used to log generations from your model.

4.3 SFT Experiment

Using the pieces above, you will now implement the full SFT procedure (Algorithm 1) to finetune the Qwen 2.5 Math 1.5B Base model on the MATH dataset. Each example in `data/sft.jsonl` consists of a formatted prompt and a target response, where the target response includes a chain-of-thought reasoning trace and the final answer. In particular, each example is a JSON element of type `{"prompt": str, "response": str}`.

In order to track the progress of your model over the course of training, you should periodically evaluate it on the MATH validation set. To get this to work, here is some starter code to initialize vLLM and to load the policy weights into the vLLM instance before every rollout phase:

```
from vllm.model_executor import set_random_seed as vllm_set_random_seed

def init_vllm(model_id: str, device: str, seed: int, gpu_memory_utilization: float = 0.85):
    """
    Start the inference process, here we use vLLM to hold a model on
    a GPU separate from the policy.
    """
    vllm_set_random_seed(seed)

    # Monkeypatch from TRL:
    # https://github.com/huggingface/trl/blob/
    #   22759c820867c8659d00082ba8cf004e963873c1/trl/trainer/grpo_trainer.py
    # Patch vLLM to make sure we can
    # (1) place the vLLM model on the desired device (world_size_patch) and
    # (2) avoid a test that is not designed for our setting (profiling_patch).
    world_size_patch = patch("torch.distributed.get_world_size", return_value=1)
    profiling_patch = patch(
        "vllm.worker.worker.Worker._assert_memory_footprint_increased_during_profiling",
        return_value=None
    )
    with world_size_patch, profiling_patch:
        return LLM(
            model=model_id,
            device=device,
            dtype=torch.bfloat16,
            enable_prefix_caching=True,
            gpu_memory_utilization=gpu_memory_utilization,
        )

def load_policy_into_vllm_instance(policy: PreTrainedModel, llm: LLM):
    """
    Copied from https://github.com/huggingface/trl/blob/
        22759c820867c8659d00082ba8cf004e963873c1/trl/trainer/grpo_trainer.py#L670.
    """
    state_dict = policy.state_dict()
    llm_model = llm.llm_engine.model_executor.driver_worker.model_runner.model
    llm_model.load_weights(state_dict.items())
```

You may find it helpful to log metrics with respect to both the train and validation steps (this will also be useful in later RL experiments). To do this in `wandb`, you can use the following code:

```python
# Setup wandb metrics
wandb.define_metric("train_step")  # the x-axis for training
wandb.define_metric("eval_step")  # the x-axis for evaluation

# everything that starts with train/ is tied to train_step
wandb.define_metric("train/*", step_metric="train_step")

# everything that starts with eval/ is tied to eval_step
wandb.define_metric("eval/*", step_metric="eval_step")
```

Lastly, we suggest that you use gradient clipping with clip value 1.0.

Problem (`sft_experiment`): Run SFT on the MATH dataset

---

1. Run SFT on the reasoning SFT examples (provided in `data/sft.jsonl`) using the Qwen 2.5 Math 1.5B base model, varying the number of unique examples for SFT in the range {128, 256, 512, 1024}, along with using the full dataset. Tune the learning rate and batch size to achieve at least 15% validation accuracy when using the full dataset.

   Deliverable: Validation accuracy curves associated with different dataset sizes.

2. Filter the reasoning SFT examples to only include examples that produce the correct answer. Run SFT on the (full) filtered dataset and report the size of the filtered dataset and the validation accuracy you achieve.

   Deliverable: Report the size of the dataset and the validation accuracy curve you achieve. Compare your findings to the previous SFT experiment.

5 Expert Iteration for MATH

In the previous section, we observed that we can improve the performance of our SFT model by filtering out bad examples from the SFT data. In this section, we will go one step further: we will apply this filtering procedure to reasoning traces we generate from our base model itself. This process is known in the literature as *expert iteration* [Anthony et al., 2017], and in the context of language models has been explored in Cobbe et al. [2021b], Zelikman et al. [2022], Dohan et al. [2022], Gulcehre et al. [2023].

Algorithm 2 Expert iteration (EI)

Input initial policy model $\pi_{\theta_{\text{init}}}$; reward function R; task questions $\mathcal{D}$

policy model $\pi_\theta \leftarrow \pi_{\theta_{\text{init}}}$
for step = 1, ..., `n_ei_steps` **do**
  Sample a batch of questions $\mathcal{D}_b$ from $\mathcal{D}$
  Set the old policy model $\pi_{\theta_{\text{old}}} \leftarrow \pi_\theta$
  Sample G outputs $\{o^{(i)}\}_{i=1}^{G} \sim \pi_{\theta_{\text{old}}}(\cdot \mid q)$ for each question $q \in \mathcal{D}_b$
  Compute rewards $\{r^{(i)}\}_{i=1}^{G}$ for each sampled output $o^{(i)}$ by running reward function $R(q, o^{(i)})$
  Filter out wrong outputs (i.e., $o^{(i)}$ with $r^{(i)} = 0$) to obtain a dataset $\mathcal{D}_{\text{sft}}$ of correct question-response pairs
  $\pi_\theta \leftarrow \text{SFT}(\pi_\theta, \mathcal{D}_{\text{sft}})$ (Algorithm 1)
end for

Output π_θ

Next, we will run expert iteration on the MATH dataset.

As a tip, you should pass a `min_tokens` value to your vLLM `SamplingParams`, which will ensure that you do not generate an empty string (which could then cause a NaN downstream depending on your implementation). This can be done with

```
sampling_min_tokens = 4
sampling_params = SamplingParams(
      temperature=sampling_temperature,
      max_tokens=sampling_max_tokens,
      min_tokens=sampling_min_tokens,
      n=G,
      seed=seed,
  )
```

As in SFT, you should use gradient clipping with clip value 1.0.

Problem (`expert_iteration_experiment`): Run expert iteration on the MATH dataset

Run expert iteration on the MATH dataset (provided at `data/MATH/train.jsonl`) using the Qwen 2.5 Math 1.5B Base model, varying the number of rollouts G per question and the number of epochs used in the SFT step, and using `n_ei_steps` $= 5$. Vary the batch size for each expert iteration step (i.e., the size of $\mathcal{D}_b$) in $\{512, 1024, 2048\}$. (You do not need to try all possible combinations of these hyperparameters. Just enough to draw conclusions about each is fine.) Log the entropy of the model's reponses over training. Make sure to have vLLM terminate generations at the second answer tag `</answer>`, as done in the SFT section.

Deliverable: Validation accuracy curves associated with different rollout configurations. Try at least 2 different rollout counts and epoch counts.

Deliverable: A model that achieves validation accuracy of at least 15% on MATH.

Deliverable: A brief 2 sentence discussion comparing to your SFT performance, as well as performance across EI steps.

Deliverable: A plot of the entropy of the model's responses over training.

6 Primer on Policy Gradients

An exciting new finding in language model research is that performing RL against verified rewards with strong base models can lead to significant improvements in their reasoning capabilities and performance [OpenAI et al., 2024, DeepSeek-AI et al., 2025]. The strongest such open reasoning models, such as DeepSeek R1 and Kimi k1.5 [Team et al., 2025], were trained using policy gradients, a powerful reinforcement learning algorithm that can optimize arbitrary reward functions.

We provide a brief introduction to policy gradients for RL on language models below. Our presentation is based closely on a couple great resources which walk through these concepts in more depth: OpenAI's Spinning Up in Deep RL [Achiam, 2018a] and Nathan Lambert's Reinforcement Learning from Human Feedback (RLHF) Book [Lambert, 2024].

6.1 Language Models as Policies

A causal language model (LM) with parameters θ defines a probability distribution over the next token $a_t \in \mathcal{V}$ given the current text prefix s_t (the state/observation). In the context of RL, we think of the next token a_t as an *action* and the current text prefix s_t as the *state*. Hence, the LM is a *categorical stochastic policy*

$$a_t \sim \pi_\theta(\cdot \mid s_t), \qquad \pi_\theta(a_t \mid s_t) = \left[\text{softmax}\big(f_\theta(s_t)\big)\right]_{a_t}. \tag{3}$$

Two primitive operations will be needed in optimizing the policy with policy gradients:

1. *Sampling from the policy*: drawing an action a_t from the categorical distribution above;
2. *Scoring the log-likelihood of an action*: evaluating $\log \pi_\theta(a_t \mid s_t)$.

Generally, when doing RL with LLMs, s_t is the partial completion/solution produced so far, and each a_t is the next token of the solution; the episode ends when an end-of-text token is emitted, like <|end_of_text|>, or </answer> in the case of our r1_zero prompt.

6.2 Trajectories

A (finite-horizon) trajectory is the interleaved sequence of states and actions experienced by an agent:

$$\tau = (s_0, a_0, s_1, a_1, \ldots, s_T, a_T)\,, \tag{4}$$

where T is the length of the trajectory, i.e., a_T is an end-of-text token or we have reached a maximum generation budget in tokens.

The initial state is drawn from the start distribution, $s_0 \sim \rho_0(s_0)$; in the case of RL with LLMs, $\rho_0(s_0)$ is a distribution over formatted prompts. In general settings, state transitions follow some environment dynamics $s_{t+1} \sim P(\cdot \mid s_t, a_t)$. In RL with LLMs, the environment is deterministic: the next state is the old prefix concatenated with the emitted token, $s_{t+1} = s_t \| a_t$. Trajectories are also called *episodes* or *rollouts*; we will use these terms interchangeably.

6.3 Rewards and Return

A scalar reward $r_t = R(s_t, a_t)$ judges the immediate quality of the action taken at state s_t. For RL on verified domains, it is standard to assign zero reward to intermediate steps and a *verified reward* to the terminal action

$$r_T = R(s_T, a_T) := \begin{cases} 1 & \text{if the trajectory } s_T \| a_T \text{ matches the ground-truth according to our reward function} \\ 0 & \text{otherwise.} \end{cases}$$

The *return* $R(\tau)$ aggregates rewards along the trajectory. Two common choices are *finite-horizon undiscounted* returns

$$R(\tau) := \sum_{t=0}^{T} r_t, \tag{5}$$

and *infinite-horizon discounted* returns

$$R(\tau) := \sum_{t=0}^{\infty} \gamma^t r_t, \qquad 0 < \gamma < 1. \tag{6}$$

In our case, we will use the undiscounted formulation since episodes have a natural termination point (end-of-text or max generation length).

The objective of the agent is to maximize the expected return

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta}\left[R(\tau)\right], \tag{7}$$

leading to the optimization problem

$$\theta^* = \arg\max_\theta J(\theta). \tag{8}$$

6.4 Vanilla Policy Gradient

Next, let us attempt to learn policy parameters θ with *gradient ascent* on the expected return:

$$\theta_{k+1} = \theta_k + \alpha \nabla_\theta J(\theta_k). \tag{9}$$

The core identity that we will use to do this is the REINFORCE policy gradient, shown below.

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{\tau \sim \pi_\theta}\left[\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t|s_t) R(\tau)\right]. \tag{10}$$

Deriving the policy gradient. How did we get this equation? For completeness, we will give a derivation of this identity below. We will make use of a few identities.

1. The probability of a trajectory is given by

$$P(\tau \mid \theta) = \rho_0(s_0) \prod_{t=0}^{T} P(s_{t+1} \mid s_t, a_t)\pi_\theta(a_t \mid s_t). \tag{11}$$

   Therefore, the log-probability of a trajectory is:

$$\log P(\tau \mid \theta) = \log \rho_0(s_0) + \sum_{t=0}^{T} \big[\log P(s_{t+1} \mid s_t, a_t) + \log \pi_\theta(a_t \mid s_t)\big]. \tag{12}$$

2. The log-derivative trick:

$$\nabla_\theta P = P \nabla_\theta \log P. \tag{13}$$

3. The environment terms are constant in θ. ρ_0, $P(\cdot \mid \cdot)$ and $R(\tau)$ do not depend on the policy parameters, so

$$\nabla_\theta \rho_0 = \nabla_\theta P = \nabla_\theta R(\tau) = 0. \tag{14}$$

Applying the facts above:

$$\begin{aligned}
\nabla_\theta J(\theta) &= \nabla_\theta \mathbb{E}_{\tau\sim\pi_\theta}[R(\tau)] && (15)\\
&= \nabla_\theta \sum_\tau P(\tau|\theta) R(\tau) && (16)\\
&= \sum_\tau \nabla_\theta P(\tau|\theta) R(\tau) && (17)\\
&= \sum_\tau P(\tau|\theta) \nabla_\theta \log P(\tau|\theta) R(\tau) && \text{(Log-derivative trick)} \quad (18)\\
&= \mathbb{E}_{\tau\sim\pi_\theta}[\nabla_\theta \log P(\tau|\theta) R(\tau)], && (19)
\end{aligned}$$

and therefore, plugging in the log-probability of a trajectory and using the fact that the environment terms are constant in θ, we get the *vanilla* or REINFORCE policy gradient:

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t|s_t) R(\tau)\right]. \tag{20}$$

Intuitively, this gradient will increase the log probability of every action in a trajectory that has high return, and decrease them otherwise.

Sample estimate of the gradient. Given a batch of N rollouts $\mathcal{D} = \{\tau^{(i)}\}_{i=1}^{N}$ collected by sampling a starting state $s_0^{(i)} \sim \rho_0(s_0)$ and then running the policy π_θ in the environment, we form an unbiased estimator of the gradient as

$$\widehat{g} = \frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t^{(i)} \mid s_t^{(i)}) R(\tau^{(i)}). \tag{21}$$

This vector is used in the gradient-ascent update $\theta \leftarrow \theta + \alpha \widehat{g}$.

6.5 Policy Gradient Baselines

The main issue with vanilla policy gradient is the high variance of the gradient estimate. A common technique to mitigate this is to subtract from the reward a *baseline* function b that depends only on the state. This is a type of *control variate* [Ross, 2022]: the idea is to decrease the variance of the estimator by subtracting a term that is correlated with it, without introducing bias.

Let us define the baselined policy gradient as:

$$B = \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{T}\nabla_\theta \log \pi_\theta(a_t|s_t)\big(R(\tau) - b(s_t)\big)\right]. \tag{22}$$

As an example, a reasonable baseline is the on-policy value function $V^\pi(s) = \mathbb{E}_{\tau\sim\pi_\theta}\left[R(\tau)|s_t = s\right]$, i.e., the expected return if we start at $s_t = s$ and follow the policy π_θ from there. Then, the quantity $(R(\tau) - V^\pi(s_t))$ is, intuitively, how much better the realized trajectory is than expected.

As long as the baseline depends only on the state, the baselined policy gradient is unbiased. We can see this by rewriting the baselined policy gradient as

$$B = \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{T}\nabla_\theta \log \pi_\theta(a_t|s_t)R(\tau)\right] - \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{T}\nabla_\theta \log \pi_\theta(a_t|s_t)b(s_t)\right]. \tag{23}$$

Focusing on the baseline term, we see that

$$\mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{T}\nabla_\theta \log \pi_\theta(a_t|s_t)b(s_t)\right] = \sum_{t=0}^{T}\mathbb{E}_{s_t}\Big[b(s_t)\mathbb{E}_{a_t\sim\pi_\theta(\cdot|s_t)}\left[\nabla_\theta \log \pi_\theta(a_t \mid s_t)\right]\Big]. \tag{24}$$

In general, the expectation of the score function is zero: $\mathbb{E}_{x\sim P_\theta}\left[\nabla_\theta \log P_\theta(x)\right] = 0$. Therefore, the expression in Eq. 24 is zero and

$$B = \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{T}\nabla_\theta \log \pi_\theta(a_t|s_t)R(\tau)\right] - 0 = \nabla_\theta J(\pi_\theta), \tag{25}$$

so we conclude that the baselined policy gradient is unbiased. We will later run an experiment to see whether baselining improves downstream performance.

A note on policy gradient "losses." When we implement policy gradient methods in a framework like PyTorch, we will define a so-called policy gradient loss `pg_loss` such that calling `pg_loss.backward()` will populate the gradient buffers of our model parameters with our approximate policy gradient $\widehat{g}$. In math, it can be stated as

$$\texttt{pg_loss} = \frac{1}{N}\sum_{i=1}^{N}\sum_{t=0}^{T}\log \pi_\theta(a_t^{(i)}|s_t^{(i)})(R(\tau^{(i)}) - b(s_t^{(i)})). \tag{26}$$

`pg_loss` is not a loss in the canonical sense—it's not meaningful to report `pg_loss` on the train or validation set as an evaluation metric, and a good validation `pg_loss` doesn't indicate that our model is generalizing well. The `pg_loss` is really just some scalar such that when we call `pg_loss.backward()`, the gradients we obtain through backprop are the approximate policy gradient $\widehat{g}$.

When doing RL, you should always **log and report train and validation rewards**. These are the "meaningful" evaluation metrics and what we are attempting to optimize with policy gradient methods.

6.6 Off-Policy Policy Gradient

REINFORCE is an *on-policy* algorithm: the training data is collected by the same policy that we are optimizing. To see this, let us write out the REINFORCE algorithm:

1. Sample a batch of rollouts $\{\tau^{(i)}\}_{i=1}^{N}$ from the current policy π_θ.

2. Approximate the policy gradient as $\nabla_\theta J(\pi_\theta) \approx \widehat{g} = \frac{1}{N}\sum_{i=1}^{N}\sum_{t=0}^{T}\nabla_\theta \log \pi_\theta(a_t^{(i)}|s_t^{(i)})R(\tau^{(i)})$.

3. Update the policy parameters using the computed gradient: $\theta \leftarrow \theta + \alpha\widehat{g}$.

We need to do a lot of inference to sample a new batch of rollouts, only to take just one gradient step. The behavior of an LM generally cannot change significantly in a single step, so this on-policy approach is highly inefficient.

Off-policy policy gradient. In off-policy learning, we instead have rollouts sampled from some policy other than the one we are optimizing. Off-policy variants of popular policy gradient algorithms like PPO and GRPO use rollouts from a previous version of the policy $\pi_{\theta_{\text{old}}}$ to optimize the current policy π_θ. The off-policy policy gradient estimate is

$$\widehat{g}_{\text{off-policy}} = \frac{1}{N}\sum_{i=1}^{N}\sum_{t=0}^{T}\frac{\pi_\theta(a_t^{(i)}|s_t^{(i)})}{\pi_{\theta_{\text{old}}}(a_t^{(i)}|s_t^{(i)})}\nabla_\theta \log \pi_\theta(a_t^{(i)}|s_t^{(i)})R(\tau^{(i)}). \tag{27}$$

This looks like an importance sampled version of the vanilla policy gradient, with reweighting terms $\frac{\pi_\theta(a_t^{(i)}|s_t^{(i)})}{\pi_{\theta_{\text{old}}}(a_t^{(i)}|s_t^{(i)})}$. Indeed, Eq. 27 can be derived by importance sampling and applying an approximation that is reasonable as long as π_θ and $\pi_{\theta_{\text{old}}}$ are not too different: see Degris et al. [2013] for more on this.

7 Group Relative Policy Optimization

Next, we will describe Group Relative Policy Optimization (GRPO), the variant of policy gradient that you will implement and experiment with for solving math problems.

7.1 GRPO Algorithm

Advantage estimation. The core idea of GRPO is to sample many outputs for each question from the policy π_θ and use them to compute a baseline. This is convenient because we avoid the need to learn a neural value function $V_\phi(s)$, which can be hard to train and is cumbersome from the systems perspective. For a question q and group outputs $\{o^{(i)}\}_{i=1}^{G} \sim \pi_\theta(\cdot|q)$, let $r^{(i)} = R(q, o^{(i)})$ be the reward for the i-th output. DeepSeekMath [Shao et al., 2024] and DeepSeek R1 [DeepSeek-AI et al., 2025] compute the group-normalized reward for the i-th output as

$$A^{(i)} = \frac{r^{(i)} - \text{mean}(r^{(1)}, r^{(2)}, \ldots, r^{(G)})}{\text{std}(r^{(1)}, r^{(2)}, \ldots, r^{(G)}) + \texttt{advantage_eps}}, \tag{28}$$

where `advantage_eps` is a small constant to prevent division by zero. Note that this *advantage* $A^{(i)}$ is the same for each token in the response, i.e., $A_t^{(i)} = A^{(i)}, \forall t \in 1, \ldots, |o^{(i)}|$, so we drop the t subscript in the following.

High-level algorithm. Before we dive into the GRPO objective, let us first get an idea of the train loop by writing out the algorithm from Shao et al. [2024] in Algorithm 3.[2]

GRPO objective. The GRPO objective combines three ideas:

1. Off-policy policy gradient, as in Eq. 27.

2. Computing advantages $A^{(i)}$ with group normalization, as in Eq. 28.

[2]This is a special case of DeepSeekMath's GRPO with a verified reward function, no KL term, and no iterative update of the reference and reward model.

Algorithm 3 Group Relative Policy Optimization (GRPO)

Input initial policy model $\pi_{\theta_{\text{init}}}$; reward function R; task questions $\mathcal{D}$

policy model $\pi_\theta \leftarrow \pi_{\theta_{\text{init}}}$
for step = 1, ..., `n_grpo_steps` **do**
  Sample a batch of questions $\mathcal{D}_b$ from $\mathcal{D}$
  Set the old policy model $\pi_{\theta_{\text{old}}} \leftarrow \pi_\theta$
  Sample G outputs $\{o^{(i)}\}_{i=1}^{G} \sim \pi_{\theta_{\text{old}}}(\cdot \mid q)$ for each question $q \in \mathcal{D}_b$
  Compute rewards $\{r^{(i)}\}_{i=1}^{G}$ for each sampled output $o^{(i)}$ by running reward function $R(q, o^{(i)})$
  Compute $A^{(i)}$ with group normalization (Eq. 28)
  for train step = 1, ..., `n_train_steps_per_rollout_batch` **do**
    Update the policy model π_θ by maximizing the GRPO-Clip objective (to be discussed, Eq. 29)
  end for
end for

Output π_θ

3. A clipping mechanism, as in Proximal Policy Optimization (PPO, Schulman et al. [2017]).

The purpose of clipping is to maintain stability when taking many gradient steps on a single batch of rollouts. It works by keeping the policy π_θ from straying too far from the old policy.

Let us first write out the full GRPO-Clip objective, and then we can build some intuition on what the clipping does:

$$J_{\text{GRPO-Clip}}(\theta) = \mathbb{E}_{q\sim\mathcal{D},\ \{o^{(i)}\}_{i=1}^{G}\sim\pi_\theta(\cdot|q)} \left[\frac{1}{G}\sum_{i=1}^{G} \frac{1}{|o^{(i)}|} \sum_{t=1}^{|o^{(i)}|} \underbrace{\min\left(\frac{\pi_\theta(o_t^{(i)} \mid q, o_{<t}^{(i)})}{\pi_{\theta_{\text{old}}}(o_t^{(i)} \mid q, o_{<t}^{(i)})} A^{(i)}, \text{clip}\left(\frac{\pi_\theta(o_t^{(i)} \mid q, o_{<t}^{(i)})}{\pi_{\theta_{\text{old}}}(o_t^{(i)} \mid q, o_{<t}^{(i)})}, 1-\epsilon,\ 1+\epsilon \right) A^{(i)} \right)}_{\text{per-token objective}} \right]. \tag{29}$$

The hyperparameter $\epsilon > 0$ controls how much the policy can change. To see this, we can rewrite the per-token objective in a more intuitive way following Achiam [2018a,b]. Define the function

$$g(\epsilon, A^{(i)}) = \begin{cases} (1+\epsilon)A^{(i)} & \text{if } A^{(i)} \geq 0 \\ (1-\epsilon)A^{(i)} & \text{if } A^{(i)} < 0. \end{cases} \tag{30}$$

We can rewrite the per-token objective as

$$\text{per-token objective} = \min\left(\frac{\pi_\theta(o_t^{(i)} \mid q, o_{<t}^{(i)})}{\pi_{\theta_{\text{old}}}(o_t^{(i)} \mid q, o_{<t}^{(i)})} A^{(i)}, g(\epsilon, A^{(i)}) \right)$$

We can now reason by cases. When the advantage $A^{(i)}$ is positive, the per-token objective simplifies to

$$\text{per-token objective} = \min\left(\frac{\pi_\theta(o_t^{(i)} \mid q, o_{<t}^{(i)})}{\pi_{\theta_{\text{old}}}(o_t^{(i)} \mid q, o_{<t}^{(i)})}, 1+\epsilon \right) A^{(i)}.$$

Since $A^{(i)} > 0$, the objective goes up if the action $o_t^{(i)}$ becomes more likely under π_θ, i.e., if $\pi_\theta(o_t^{(i)} \mid q, o_{<t}^{(i)})$ increases. The clipping with min limits how much the objective can increase: once $\pi_\theta(o_t^{(i)} \mid q, o_{<t}^{(i)}) > (1+\epsilon)\pi_{\theta_{\text{old}}}(o_t^{(i)} \mid q, o_{<t}^{(i)})$, this per-token objective hits its maximum value of $(1+\epsilon)A^{(i)}$. So, the policy π_θ is not incentivized to go very far from the old policy $\pi_{\theta_{\text{old}}}$.

Analogously, when the advantage $A^{(i)}$ is negative, the model tries to drive down $\pi_\theta(o_t^{(i)} \mid q, o_{<t}^{(i)})$, but is not incentivized to decrease it below $(1-\epsilon)\pi_{\theta_{\text{old}}}(o_t^{(i)} \mid q, o_{<t}^{(i)})$ (refer to Achiam [2018b] for the full argument).

7.2 Implementation

Now that we have a high-level understanding of the GRPO training loop and objective, we will start implementing pieces of it. Many of the pieces implemented in the SFT and EI sections will also be reused for GRPO.

Computing advantages (group-normalized rewards). First, we will implement the logic to compute advantages for each example in a rollout batch, i.e., the group-normalized rewards. We will consider two possible ways to obtain group-normalized rewards: the approach presented above in Eq. 28, and a recent simplified approach.

Dr. GRPO [Liu et al., 2025] highlights that normalizing by $\text{std}(r^{(1)}, r^{(2)}, \ldots, r^{(G)})$ rewards questions in a batch with low variation in answer correctness, which may not be desirable. They propose simply removing the normalization step, computing

$$A^{(i)} = r^{(i)} - \text{mean}(r^{(1)}, r^{(2)}, \ldots, r^{(G)}). \tag{31}$$

We will implement both variants and compare their performance later in the assignment.

Problem (`compute_group_normalized_rewards`): Group normalization

Deliverable: Implement a method `compute_group_normalized_rewards` that calculates raw rewards for each rollout response, normalizes them within their groups, and returns both the normalized and raw rewards along with any metadata you think is useful.
The following interface is recommended:

```
def compute_group_normalized_rewards(
    reward_fn,
    rollout_responses,
    repeated_ground_truths,
    group_size,
    advantage_eps,
    normalize_by_std,
):
```

Compute rewards for each group of rollout responses, normalized by the group size.

Args:

`reward_fn: Callable[[str, str], dict[str, float]]` Scores the rollout responses against the ground truths, producing a dict with keys `"reward"`, `"format_reward"`, and `"answer_reward"`.

`rollout_responses: list[str]` Rollouts from the policy. The length of this list is `rollout_batch_size = n_prompts_per_rollout_batch * group_size`.

`repeated_ground_truths: list[str]` The ground truths for the examples. The length of this list is `rollout_batch_size`, because the ground truth for each example is repeated `group_size` times.

`group_size: int` Number of responses per question (group).

`advantage_eps: float` Small constant to avoid division by zero in normalization.

`normalize_by_std: bool` If `True`, divide by the per-group standard deviation; otherwise subtract only the group mean.

Returns:

`tuple[torch.Tensor, torch.Tensor, dict[str, float]]`.

`advantages` shape `(rollout_batch_size,)`. Group-normalized rewards for each rollout response.

`raw_rewards` shape `(rollout_batch_size,)`. Unnormalized rewards for each rollout response.

`metadata` your choice of other statistics to log (e.g. mean, std, max/min of rewards).

To test your code, implement `[adapters.run_compute_group_normalized_rewards]`. Then, run the test with `uv run pytest -k test_compute_group_normalized_rewards` and make sure your implementation passes it.

Naive policy gradient loss. Next, we will implement some methods for computing "losses".

As a **reminder/disclaimer**, these are not really losses in the canonical sense and should not be reported as evaluation metrics. When it comes to RL, you should instead track the train and validation returns, among other metrics (cf. Section 6.5 for discussion).

We will start with the naive policy gradient loss, which simply multiplies the advantage by the log-probability of actions (and negates). With question q, response o, and response token o_t, the naive per-token policy gradient loss is

$$-A_t \cdot \log p_\theta(o_t | q, o_{<t}). \tag{32}$$

Problem (`compute_naive_policy_gradient_loss`): Naive policy gradient

Deliverable: Implement a method `compute_naive_policy_gradient_loss` that computes the per-token policy-gradient loss using raw rewards or pre-computed advantages.
The following interface is recommended:

```
def compute_naive_policy_gradient_loss(
    raw_rewards_or_advantages: torch.Tensor,
    policy_log_probs: torch.Tensor,
) -> torch.Tensor:
```

Compute the policy-gradient loss at every token, where `raw_rewards_or_advantages` is either the raw reward or an already-normalized advantage.

Args:

`raw_rewards_or_advantages: torch.Tensor` Shape `(batch_size, 1)`, scalar reward/advantage for each rollout response.

`policy_log_probs: torch.Tensor` Shape `(batch_size, sequence_length)`, logprobs for each token.

Returns:

`torch.Tensor` Shape `(batch_size, sequence_length)`, the per-token policy-gradient loss (to be aggregated across the batch and sequence dimensions in the training loop).

Implementation tips:

- Broadcast the `raw_rewards_or_advantages` over the `sequence_length` dimension.

To test your code, implement [adapters.run_compute_naive_policy_gradient_loss]. Then run `uv run pytest -k test_compute_naive_policy_gradient_loss` and ensure the test passes.

GRPO-Clip loss. Next, we will implement the more interesting GRPO-Clip loss.

The per-token GRPO-Clip loss is

$$-\min\left(\frac{\pi_\theta(o_t|q,o_{<t})}{\pi_{\theta_{\text{old}}}(o_t|q,o_{<t})}A_t, \text{clip}\left(\frac{\pi_\theta(o_t|q,o_{<t})}{\pi_{\theta_{\text{old}}}(o_t|q,o_{<t})}, 1-\epsilon, 1+\epsilon\right)A_t\right). \tag{33}$$

Problem (`compute_grpo_clip_loss`): GRPO-Clip loss

Deliverable: Implement a method `compute_grpo_clip_loss` that computes the per-token GRPO-Clip loss.
The following interface is recommended:

```
def compute_grpo_clip_loss(
    advantages: torch.Tensor,
    policy_log_probs: torch.Tensor,
    old_log_probs: torch.Tensor,
    cliprange: float,
) -> tuple[torch.Tensor, dict[str, torch.Tensor]]:
```

Args:

`advantages: torch.Tensor` Shape `(batch_size, 1)`, per-example advantages A.

`policy_log_probs: torch.Tensor` Shape `(batch_size, sequence_length)`, per-token log probs from the policy being trained.

`old_log_probs: torch.Tensor` Shape `(batch_size, sequence_length)`, per-token log probs from the old policy.

`cliprange: float` Clip parameter ϵ (e.g. 0.2).

Returns:

`tuple[torch.Tensor, dict[str, torch.Tensor]]`.

loss `torch.Tensor` of shape `(batch_size, sequence_length)`, the per-token clipped loss.

metadata dict containing whatever you want to log. We suggest logging whether each token was clipped or not, i.e., whether the clipped policy gradient loss on the RHS of the min was lower than the LHS.

Implementation tips:

- Broadcast `advantages` over `sequence_length`.

To test your code, implement [adapters.run_compute_grpo_clip_loss]. Then run `uv run pytest -k test_compute_grpo_clip_loss` and ensure the test passes.

Policy gradient loss wrapper. We will be running ablations comparing three different versions of policy gradient:

(a) `no_baseline`: Naive policy gradient loss without a baseline, i.e., advantage is just the raw rewards $A = R(q, o)$.

(b) `reinforce_with_baseline`: Naive policy gradient loss but using our group-normalized rewards as the advantage. If $\bar{r}$ are the group-normalized rewards from `compute_group_normalized_rewards` (which may or may not be normalized by the group standard deviation), then $A = \bar{r}$.

(c) `grpo_clip`: GRPO-Clip loss.

For convenience, we will implement a wrapper that lets us easily swap between these three policy gradient losses.

Problem (`compute_policy_gradient_loss`): Policy-gradient wrapper

Deliverable: Implement `compute_policy_gradient_loss`, a convenience wrapper that dispatches to the correct loss routine (`no_baseline`, `reinforce_with_baseline`, or `grpo_clip`) and returns both the per-token loss and any auxiliary statistics.
The following interface is recommended:

```python
def compute_policy_gradient_loss(
    policy_log_probs: torch.Tensor,
    loss_type: Literal["no_baseline", "reinforce_with_baseline", "grpo_clip"],
    raw_rewards: torch.Tensor | None = None,
    advantages: torch.Tensor | None = None,
    old_log_probs: torch.Tensor | None = None,
    cliprange: float | None = None,
) -> tuple[torch.Tensor, dict[str, torch.Tensor]]:
```

Select and compute the desired policy-gradient loss.

Args:

`policy_log_probs` `(batch_size, sequence_length)`, per-token log-probabilities from the policy being trained.

`loss_type` One of `"no_baseline"`, `"reinforce_with_baseline"`, or `"grpo_clip"`.

`raw_rewards` Required if `loss_type == "no_baseline"`; shape `(batch_size, 1)`.

`advantages` Required for `"reinforce_with_baseline"` and `"grpo_clip"`; shape `(batch_size, 1)`.

`old_log_probs` Required for `"grpo_clip"`; shape `(batch_size, sequence_length)`.

`cliprange` Required for `"grpo_clip"`; scalar ϵ used for clipping.

Returns:

`tuple[torch.Tensor, dict[str, torch.Tensor]]`.

loss `(batch_size, sequence_length)`, per-token loss.
metadata dict, statistics from the underlying routine (e.g., clip fraction for GRPO-Clip).

Implementation tips:

- Delegate to `compute_naive_policy_gradient_loss` or `compute_grpo_clip_loss`.
- Perform argument checks (see assertion pattern above).
- Aggregate any returned metadata into a single dict.

To test your code, implement `[adapters.run_compute_policy_gradient_loss]`. Then run `uv run pytest -k test_compute_policy_gradient_loss` and verify it passes.

Masked mean. Up to this point, we have the logic needed to compute advantages, log probabilities, per-token losses, and helpful statistics like per-token entropies and clip fractions. To reduce our per-token loss tensors of shape `(batch_size, sequence_length)` to a vector of losses (one scalar for each example), we will compute the mean of the loss over the sequence dimension, but only over the indices corresponding to the response (i.e., the token positions for which `mask[i, j]==1`).

Normalizing by the sequence length has been canonical in most codebases for doing RL with LLMs, but it is not obvious that this is the right thing to do—you may notice, looking at our statement of the policy gradient estimate in (21), that there is no normalization factor $\frac{1}{T^{(i)}}$. We will start with this standard primitive, often referred to as a `masked_mean`, but will later test out using the `masked_normalize` method that we implemented during SFT.

We will allow specification of the dimension over which we compute the mean, and if `dim` is `None`, we will compute the mean over all masked elements. This may be useful to obtain average per-token entropies on the response tokens, clip fractions, etc.

Problem (`masked_mean`): Masked mean

Deliverable: Implement a method `masked_mean` that averages tensor elements while respecting a boolean mask.
The following interface is recommended:

```
def masked_mean(
    tensor: torch.Tensor,
    mask: torch.Tensor,
    dim: int | None = None,
) -> torch.Tensor:
```

Compute the mean of `tensor` along a given dimension, considering only those elements where `mask == 1`.

Args:

`tensor: torch.Tensor` The data to be averaged.

`mask: torch.Tensor` Same shape as `tensor`; positions with `1` are included in the mean.

`dim: int | None` Dimension over which to average. If `None`, compute the mean over all masked elements.

Returns:

`torch.Tensor` The masked mean; shape matches `tensor.mean(dim)` semantics.

To test your code, implement `[adapters.run_masked_mean]`. Then run `uv run pytest -k test_masked_mean` and ensure it passes.

GRPO microbatch train step. Now we are ready to implement a single microbatch train step for GRPO (recall that for a train minibatch, we iterate over many microbatches if `gradient_accumulation_steps > 1`).

Specifically, given the raw rewards or advantages and log probs, we will compute the per-token loss, use `masked_mean` to aggregate to a scalar loss per example, average over the batch dimension, adjust for gradient accumulation, and backpropagate.

Problem (`grpo_microbatch_train_step`): Microbatch train step

Deliverable: Implement a single micro-batch update for GRPO, including policy-gradient loss, averaging with a mask, and gradient scaling.
The following interface is recommended:

```python
def grpo_microbatch_train_step(
    policy_log_probs: torch.Tensor,
    response_mask: torch.Tensor,
    gradient_accumulation_steps: int,
    loss_type: Literal["no_baseline", "reinforce_with_baseline", "grpo_clip"],
    raw_rewards: torch.Tensor | None = None,
    advantages: torch.Tensor | None = None,
    old_log_probs: torch.Tensor | None = None,
    cliprange: float | None = None,
) -> tuple[torch.Tensor, dict[str, torch.Tensor]]:
```

Execute a forward-and-backward pass on a microbatch.

Args:

`policy_log_probs` `(batch_size, sequence_length)`, per-token log-probabilities from the policy being trained.

`response_mask` `(batch_size, sequence_length)`, `1` for response tokens, `0` for prompt/padding.

`gradient_accumulation_steps` Number of microbatches per optimizer step.

`loss_type` One of `"no_baseline"`, `"reinforce_with_baseline"`, `"grpo_clip"`.

`raw_rewards` Needed when `loss_type == "no_baseline"`; shape `(batch_size, 1)`.

`advantages` Needed when `loss_type != "no_baseline"`; shape `(batch_size, 1)`.

`old_log_probs` Required for GRPO-Clip; shape `(batch_size, sequence_length)`.

`cliprange` Clip parameter ϵ for GRPO-Clip.

Returns:

`tuple[torch.Tensor, dict[str, torch.Tensor]]`.

loss scalar tensor. The microbatch loss, adjusted for gradient accumulation. We return this so we can log it.

metadata Dict with metadata from the underlying loss call, and any other statistics you might want to log.

Implementation tips:

- You should call `loss.backward()` in this function. Make sure to adjust for gradient accumulation.

To test your code, implement `[adapters.run_grpo_microbatch_train_step]`. Then run `uv run pytest -k test_grpo_microbatch_train_step` and confirm it passes.

Putting it all together: GRPO train loop. Now we will put together a complete train loop for GRPO. You should refer to the algorithm in Section 7.1 for the overall structure, using the methods we've implemented where appropriate.

Below we provide some starter hyperparameters. If you have a correct implementation, you should see reasonable results with these.

```
n_grpo_steps: int = 200
learning_rate: float = 1e-5
advantage_eps: float = 1e-6
rollout_batch_size: int = 256
group_size: int = 8
sampling_temperature: float = 1.0
sampling_min_tokens: int = 4  # As in Expiter, disallow empty string responses
sampling_max_tokens: int = 1024
epochs_per_rollout_batch: int = 1  # On-policy
train_batch_size: int = 256  # On-policy
gradient_accumulation_steps: int = 128  # microbatch size is 2, will fit on H100
gpu_memory_utilization: float = 0.85
loss_type: Literal[
    "no_baseline",
    "reinforce_with_baseline",
    "grpo_clip",
] = "reinforce_with_baseline"
use_std_normalization: bool = True
optimizer = torch.optim.AdamW(
    policy.parameters(),
    lr=learning_rate,
    weight_decay=0.0,
    betas=(0.9, 0.95),
)
```

These default hyperparameters will start you in the on-policy setting—for each rollout batch, we take a single gradient step. In terms of hyperparameters, this means that `train_batch_size` is equal to `rollout_batch_size`, and `epochs_per_rollout_batch` is equal to 1.

Here are some sanity check asserts and constants that should remove some edge cases and point you in the right direction:

```
assert train_batch_size % gradient_accumulation_steps == 0, (
    "train_batch_size must be divisible by gradient_accumulation_steps"
)
micro_train_batch_size = train_batch_size // gradient_accumulation_steps
assert rollout_batch_size % group_size == 0, (
    "rollout_batch_size must be divisible by group_size"
)
n_prompts_per_rollout_batch = rollout_batch_size // group_size
assert train_batch_size >= group_size, (
    "train_batch_size must be greater than or equal to group_size"
)
n_microbatches_per_rollout_batch = rollout_batch_size // micro_train_batch_size
```

And here are a few additional tips:

- Remember to use the `r1_zero` prompt, and direct vLLM to stop generation at the second answer tag `</answer>`, as in the previous experiments.
- We suggest using `typer` for argument parsing.
- Use gradient clipping with clip value 1.0.
- You should routinely log validation rewards (e.g., every 5 or 10 steps). You should evaluate on at least 1024 validation examples to compare hyperparameters, as CoT/RL evaluations can be noisy.
- With our implementation of the losses, GRPO-Clip should only be used when off-policy (since it requires the old log-probabilities).

- In the off-policy setting with multiple epochs of gradient updates per rollout batch, it would be wasteful to recompute the old log-probabilities for each epoch. Instead, we can compute the old log-probabilities once and reuse them for each epoch.
- You should not differentiate with respect to the old log-probabilities.
- You should log some or all of the following for each optimizer update:
  - The loss.
  - Gradient norm.
  - Token entropy.
  - Clip fraction, if off-policy.
  - Train rewards (total, format, and answer).
  - Anything else you think could be useful for debugging.

Problem (`grpo_train_loop`): GRPO train loop

Deliverable: Implement a complete train loop for GRPO. Begin training a policy on MATH and confirm that you see validation rewards improving, along with sensible rollouts over time. Provide a plot with the validation rewards with respect to steps, and a few example rollouts over time.

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